FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Late Pregnancy Haemorrhage / Antepartum Haemorrhage

Lecture | Week 5 · Bleeding, cancer & fertility | Obstetrics | Dr Kavuma

Antepartum haemorrhage is bleeding from the genital tract after the threshold of viability (28 weeks in Uganda, 24 weeks in UK guidance) and before the birth of the baby. The big three are placenta praevia (painless, soft uterus), abruptio placentae (painful, tense uterus, may be concealed, DIC) and uterine rupture, and no digital vaginal examination is done until praevia has been excluded.

Must know before the ward round

1. APH = bleeding from the genital tract after the period of viability (Uganda: 28 weeks; RCOG: 24+0 weeks) and before delivery of the baby.
2. Never do a digital vaginal examination in APH until placenta praevia is excluded by ultrasound (or in theatre with everything ready for CS).
3. Praevia: painless, causeless, recurrent bright-red bleeding; soft non-tender uterus; high head or malpresentation; fetus usually fine.
4. Abruption: painful bleeding, tense 'woody' tender uterus, shock out of proportion to visible loss, fetal distress or death; watch for DIC and renal failure.
5. Shock out of proportion to visible loss = concealed abruption or rupture: calculate the shock index (≥0.9 abnormal), mark the fundus and re-measure.
6. Abruption is a clinical diagnosis: a normal ultrasound does not exclude it. Do a 20-minute clotting test on arrival and repeat every 2–4 hours.
7. Low-lying placenta = edge <20 mm from the internal os on ultrasound after 16 weeks; praevia = covers the os. Transvaginal scan is safe and most accurate; rescan at 32 weeks.
8. Abruption with a dead fetus: resuscitate, correct coagulopathy, and aim for vaginal delivery (ARM + oxytocin). Abruption with a live, distressed, viable fetus: urgent CS.
9. Planned CS: uncomplicated praevia at 36+0–37+0 weeks; suspected PAS at 35+0–36+6 weeks in a tertiary centre with 4–6 units of blood and consent for hysterectomy.
10. Vasa praevia triad: rupture of membranes, painless bleeding, then acute fetal distress. It is fetal blood, so the baby exsanguinates quickly; emergency CS.
11. Previous CS plus anterior low-lying placenta or praevia = suspect placenta accreta spectrum (about 3%, 11%, 40% and over 60% with 0, 1, 2 and 3+ previous CS).
12. Give steroids (dexamethasone 6 mg IM 12-hourly × 4) if delivery is likely before 34 weeks, and anti-D within 72 h if Rh-negative.

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1. Definition

Antepartum haemorrhage (APH) is bleeding from or into the genital tract **after the period of viability** and before the baby is born. It complicates about **2–5% of pregnancies** and is a major cause of **maternal death, stillbirth and preterm birth**.

Term	Definition
APH	Bleeding from or into the genital tract from **28 weeks** (Uganda, where 28 weeks is the local threshold of viability) or **24+0 weeks** (RCOG) until the birth of the baby
Placenta praevia	The placenta is implanted wholly or partly in the **lower uterine segment**. On modern ultrasound: **praevia** = covers the internal os; **low-lying** = edge **<20 mm** from the os (after 16 weeks)
Placental abruption (abruptio placentae)	**Premature separation of a normally sited placenta** before the birth of the fetus
Vasa praevia	Unprotected fetal vessels run in the membranes **over or near the internal os**
Uterine rupture	Full-thickness tear of the uterine wall (including serosa); **dehiscence** = scar separation with serosa intact
Placenta accreta spectrum (PAS)	Abnormally adherent placenta: **accreta** (onto the myometrium), **increta** (into it), **percreta** (through the serosa, possibly into the bladder)

The three life-threatening causes you must identify quickly are **placenta praevia, abruptio placentae and uterine rupture**. Everything else (cervical lesions, infections, vasa praevia, show, trauma) comes after those are excluded.

RED FLAG

No digital vaginal examination (VE) until placenta praevia has been excluded. A finger through the cervix into a praevia can provoke catastrophic haemorrhage. First: resuscitate, then do an abdominal examination and an **ultrasound**. A gentle **speculum** examination is allowed once praevia is excluded, or cautiously to see the cervix.

CLINICAL PEARL

The one thing never to forget: **the amount of blood you see is not the amount she has lost.** In abruption much of the blood is hidden behind the placenta and inside the myometrium.

2. Epidemiology

Cause	Approximate frequency
Abruption	~1% of pregnancies (about 30% of APH)
Placenta praevia	~0.3–0.5% at term (about 1 in 200); about 20% of APH
Vasa praevia	1 in 1,200 – 1 in 5,000
Uterine rupture	0.2–0.7% of trials of labour after one CS; much higher in **obstructed labour** in unscarred multiparous uteri in low-resource settings
"Unexplained" / local causes	Up to 50% of APH episodes

UGANDA CONTEXT

In Uganda APH is managed against a background of **late ANC booking**, **limited ultrasound** (especially at HC III/IV, where the anomaly scan with placental localisation is often never done), **shortage of blood**, and **high rates of hypertension** (the leading risk factor for abruption). **Uterine rupture from neglected obstructed labour**, sometimes with herbal uterotonics (e.g. *kyogero* concoctions) or misused oxytocin/misoprostol, and increasingly from **previous CS scars**, is a real and preventable killer. A study at Mulago found that severe abruption was strongly associated with hypertension, low socio-economic status and poor ANC attendance. The MoH (Safe Mama and Essential Maternal & Newborn Care guidelines) and UCG 2023 emphasise quick identification of the big three and **referral to a CEmONC facility**.

3. Aetiology and Risk Factors

Causes of APH: placental (praevia, abruption, vasa praevia), uterine (rupture), local (cervical/vaginal lesions, infection, trauma, cancer), and unexplained.

Placenta praevia	Abruption	Uterine rupture	Vasa praevia
Previous CS (and other uterine scars: myomectomy, D&C)	**Hypertension / pre-eclampsia** (most important)	**Previous CS** (especially classical/T-incision), myomectomy	**Velamentous cord insertion**
Multiparity, maternal age >35	**Previous abruption** (recurrence about 10%)	**Obstructed labour** (grand multipara, CPD, malpresentation)	Bilobed or succenturiate placenta
Multiple pregnancy (large placenta)	**Trauma** (road accident, assault, falls)	Oxytocin/misoprostol misuse, herbal uterotonics	Low-lying placenta in the 2nd trimester
Previous praevia	Sudden uterine decompression (ARM with polyhydramnios, after first twin)	Grand multiparity	IVF pregnancy
Smoking, IVF	PPROM, chorioamnionitis	Instrumental/destructive manipulation, internal podalic version	Multiple pregnancy
Uterine anomalies, fibroids	Smoking, cocaine, thrombophilia, advanced age, multiparity, short cord	Fundal pressure, trauma	

4. Pathophysiology

Placenta praevia: why painless and why recurrent

- The placenta implants low, possibly because a scarred or poorly vascularised upper segment (previous CS, multiparity) or a very large placenta (twins) pushes implantation into the lower segment.
- From about 28 weeks the **lower segment forms and stretches** (and later effaces and dilates). The inelastic placenta can't stretch with it, so **it shears off its attachment, opening maternal sinuses**. That's why bleeding is **painless** (no retroplacental clot irritating myometrium), **"causeless"** (triggered by the normal growth of the lower segment, coitus or VE), and **recurrent** (each stretch shears a little more).
- The **lower segment has few muscle fibres**, so it contracts poorly. This is why praevia also causes **PPH after delivery**, even after a CS.
- The bleeding is **maternal blood**, so the fetus is usually fine unless the mother is in shock.
- A placenta in the lower segment **takes up space** in the pelvis, so the head stays **high and unengaged** and malpresentations (breech, transverse lie) are common.

- **PAS**: where the placenta implants on a **CS scar**, the decidua is deficient and villi invade myometrium. The risk of accreta with praevia rises with each previous CS: roughly **3% with no previous CS, 11% after 1 previous CS, 40% after 2, and over 60% after 3 or more** (Silver 2006: 3%, 11%, 40%, 61% and 67% at the first, second, third, fourth and fifth-or-more CS).

Placental abruption: why painful, why shock, why DIC

- The basic lesion is **haemorrhage into the decidua basalis**, often from diseased spiral arteries (**hypertension, pre-eclampsia**, smoking) or from shearing (trauma, sudden decompression). The haematoma dissects and separates more placenta, a **self-propagating** process.
- Blood behind the placenta **irritates the myometrium**, causing **pain, tenderness and increased tone (the "woody-hard" uterus)** with frequent small contractions.
- **Concealed bleeding**: blood trapped behind the placenta or tracking into the myometrium. Visible loss is small but the mother is **shocked**. When blood extravasates through the myometrium to the serosa, the uterus looks bruised purple-blue. This is the **Couvelaire uterus (uteroplacental apoplexy)**. It usually still contracts and is **not an indication for hysterectomy** in itself.
- **Fetus**: loss of placental surface → hypoxia → distress → death. Fetal death usually indicates **more than 50% separation** and substantial maternal blood loss (often more than 1.5 L).
- **DIC**: the damaged decidua and retroplacental clot release **thromboplastin (tissue factor)** into the maternal circulation. This triggers widespread clotting that **consumes fibrinogen, platelets and factors**, followed by fibrinolysis. It happens in about **10–20% of severe abruptions**, especially with fetal death. Hence non-clotting blood, oozing, and PPH.
- **Kidneys**: hypovolaemia plus DIC microthrombi → **acute tubular necrosis** and, in severe cases, **bilateral renal cortical necrosis**.
- **Pituitary**: shock → **Sheehan's syndrome**.

Vasa praevia: why the baby dies so fast. The vessels are **fetal**, unprotected by Wharton's jelly or placenta. When the membranes rupture (spontaneously or at ARM), a vessel tears, and the baby, whose **total blood volume is only about 80–100 mL/kg (about 250–300 mL at term)**, can **exsanguinate within minutes**. The mother is well; the **CTG shows bradycardia or a sinusoidal pattern**. Perinatal mortality is **at least 60% when diagnosed in labour**, but survival is **over 95%** with antenatal diagnosis and planned CS.

Uterine rupture

- **Scarred uterus**: the previous CS scar gives way, often insidiously, during labour (especially with **induction or augmentation**). Classical or T-shaped scars are far weaker than lower-segment transverse scars.
- **Unscarred uterus (obstructed labour)**: the upper segment keeps contracting and retracting, pulling the lower segment progressively thinner. A visible ridge forms, **Bandl's (pathological retraction) ring**, rising towards the umbilicus. Eventually the thin lower segment tears, often laterally into the broad ligament, and sometimes into the bladder.
- After rupture, the **contractions stop**, the presenting part **recedes**, the fetus is expelled into the abdomen (fetal parts easily felt), and the mother bleeds internally and externally.

Late pregnancy bleeding	fetus	high
↓		
Praevia: placenta in lower segment	Abruption: decidual haemorrhage behind a normal placenta	Rupture: scar gives way or obstructed lower segment tears
Vasa praevia: fetal vessels over the os tear at ROM		
↓		
Lower segment stretches, placenta shears off: painless recurrent bright bleeding	Retroplacental clot irritates myometrium: pain, woody uterus	Concealed blood: shock out of proportion
Thromboplastin release: DIC	Loss of placental surface: fetal distress or death	Contractions stop, fetus into abdomen, internal bleeding, shock
Fetal exsanguination: bradycardia, sinusoidal CTG, mother well		
↓		
B2	Hypovolaemia: AKI, Sheehan syndrome	PPH, non-clotting blood

Feature links

- Lower segment stretching → placental shearing → painless, causeless, recurrent bright-red bleeding (praevia).
- Placenta occupying the lower segment → high, unengaged head, breech or transverse lie (praevia).
- Posterior praevia over the sacral promontory → reduced AP diameter, cord compression → **Stallworthy's sign** (FHR slows when the head is pressed into the pelvis).
- Poorly muscular lower segment → poor contraction after delivery → PPH (praevia).
- Retroplacental clot irritating the myometrium → constant pain, tenderness, tense "woody" uterus, frequent small contractions (abruption).
- Blood trapped behind the placenta → small visible loss with tachycardia, high shock index, rising fundal height (concealed abruption).
- Extravasation into myometrium → bruised purple Couvelaire uterus.
- Loss of >50% of placental surface → fetal death; maternal loss often >1.5 L.
- Thromboplastin entering the circulation → consumption of fibrinogen and platelets → non-clotting blood, oozing, haematuria, PPH.
- Hypovolaemia + microthrombi → oliguria (<30 mL/h), ATN, renal cortical necrosis.
- Shock → pituitary infarction → Sheehan's syndrome.
- Ruptured fetal vessel → fetal exsanguination → acute bradycardia or sinusoidal trace with a well mother (vasa praevia).
- Rupture → contractions cease, presenting part recedes, fetal parts easily felt, haematuria (bladder involvement), absent FHR.

5. Classification / Staging

RCOG severity grading of APH (by the amount of blood lost)

Grade	Blood loss
Spotting	Staining or streaking on underwear/pad
Minor	<50 mL, settled
Major	50–1000 mL, no clinical shock
Massive	>1000 mL **and/or signs of shock**

Traditional classification of placenta praevia (still used in Ugandan exams and theatre notes)

Type	Description	Old term
I	Encroaches on the lower segment but doesn't reach the os	Lateral / low-lying (minor)
II	Reaches the edge of the internal os	Marginal (minor). **Type II posterior is "dangerous"**: it lies over the sacral promontory, reducing the AP diameter and compressing the cord
III	Covers the os when closed but not when dilated	Partial / incomplete central (major)
IV	Covers the os completely, even when dilated	Complete / central (major)

Abruption types: revealed (blood tracks down and out, about 65–80%), **concealed** (blood trapped behind the placenta, about 20–35%, and the most dangerous), **mixed**.

Clinical grading of abruption (Page/Sher, simplified)

Grade	Features
0	Asymptomatic; retroplacental clot found after delivery
1 (mild)	Vaginal bleeding, mild tenderness, mother and fetus well
2 (moderate)	Tender tense uterus, **fetal distress**, mother not in shock
3 (severe)	**Fetal death**, maternal shock ± **coagulopathy (3b)**

Shock index (HR ÷ SBP): **≥0.9 abnormal, ≥1.4 urgent**.

6. Clinical Features

History

Question	Why it matters
Gestational age (LNMP, earliest scan)	Viable? Steroids? Timing of delivery
Amount of blood (pads soaked, clots), colour (bright vs dark), timing, recurrence	Severity (RCOG grade); praevia bleeds are bright and recurrent; abruption blood is often dark
Pain? None, constant, colicky, back pain, sudden tearing then relief	**Painless = praevia**; **constant pain plus tense uterus = abruption**; colicky = labour; sudden severe pain then relief = rupture
Fetal movements	Reduced or absent = fetal compromise or death (abruption, vasa praevia)

Question	Why it matters
Trigger: coitus, trauma (boda-boda, road accident, assault, domestic violence), rupture of membranes, labour	Postcoital: praevia or cervical lesion; trauma: abruption; bleeding right after ROM: **vasa praevia**
Previous scans: placental site at the anomaly scan?	Known praevia or low-lying placenta changes everything
Previous CS, number and type; myomectomy, other uterine surgery	Praevia, **PAS**, rupture
Headache, visual symptoms, known high BP	Pre-eclampsia behind an abruption
In labour: duration, oxytocin, misoprostol, **herbal medicine**	Rupture
Previous APH or abruption, parity, twins	Recurrence and risk factors
Blood group/Rhesus, last Hb, HIV status, sickle cell	Anti-D, transfusion planning, PMTCT
Symptoms of shock: dizziness, thirst, fainting	Severity
Vaginal discharge, itch, postcoital bleeding, previous Pap/VIA	Local causes (cervicitis, polyp, **cervical cancer**)

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), with the obstetric abdominal examination. In heavy bleeding, resuscitation (section 9) comes first.

Introduction, consent and preparation

- Introduce yourself, confirm identity and gestation, explain, gain consent; chaperone.
- **Hand hygiene**.
- **Position and exposure**: **left lateral tilt** (avoid aortocaval compression); expose the abdomen; look at the pads and clots yourself.
- Say it out loud and write it: **"No digital vaginal examination until praevia is excluded."**

General inspection and vital signs

Sign	Mechanism	What it means
Pallor, sweating, restlessness	Hypovolaemia	Significant loss
Pads: amount, fresh or dark, clots, still flowing	Visible loss	Severity; dark blood suggests abruption
Oedema	Capillary leak	Pre-eclampsia → abruption

Vital signs: pulse, BP, RR, SpO_2, temperature, **shock index** (≥0.9 abnormal, ≥1.4 urgent); **urine output** once catheterised. A "normal" BP in a woman with pre-eclampsia may already be low for her. **Check urine for protein.**

Hands, face and neck

- Peripheral perfusion: cool hands, capillary refill (shock).
- Conjunctival pallor (anaemia).
- Oozing from cannula sites, bruising (DIC).
- Facial oedema (pre-eclampsia).

Obstetric abdominal examination

Step	Finding	What it means
Inspection	Scars; distension; Bandl's ring rising towards the umbilicus	Previous CS; obstructed labour, impending rupture

Step	Finding	What it means
Palpation: tone and tenderness (between contractions)	**Soft, relaxed, non-tender**	Praevia
	Tense, tender, "woody"; frequent small contractions	Abruption
	Scar tenderness; contractions stopped	Rupture
Fundal height	Larger than dates or **rising** against your pen mark	**Concealed haemorrhage**: mark the fundus with a pen and re-measure
Lie and presentation	**High head / malpresentation** (breech, transverse)	Praevia
	Fetal parts easily palpable, uterus felt separately	Rupture
Engagement	Unengaged at term	Praevia
Auscultation of the fetal heart (Pinard, Doppler, CTG)	Absent	Severe abruption (anticipate DIC)
	Bradycardia or sinusoidal pattern	Vasa praevia
	Distress	Abruption

> NOTE
> **Stallworthy's sign:** slowing of the fetal heart when the head is pressed into the pelvis, recovering on release. It suggests a posterior praevia. It's a classic viva sign, though rarely used today.

Also look for a **distended bladder and haematuria** (rupture in obstructed labour).

Completing the examination

- **Speculum examination** (only once praevia is excluded, or very gently to see the cervix if no scan is available and bleeding is minor): source of bleeding (cervical os vs a lesion), cervical dilatation, polyp, ectropion, **cervical cancer**, vaginal trauma, liquor.
- **No digital VE** until praevia has been excluded (or as a double set-up in theatre).
- Thank her, hand hygiene, summarise.

Document these positives	And these negatives
Estimated/weighed loss and when it started	"No digital VE performed"
Uterine tone, tenderness, fundal height (with time)	"Uterus soft and non-tender" (or not)
Lie, presentation, FHR	"No signs of rupture"
BP, urine protein	"Clotting test: clot formed at __ min"

Red flags

RED FLAG

- **Shock** (pulse >100, SBP <90, shock index ≥0.9), especially **out of proportion to the visible bleeding**: concealed abruption or rupture
- **Tense, tender uterus** or rising fundal height
- **Absent fetal heart**: severe abruption, anticipate DIC
- **Non-clotting blood**, oozing from sites, haematuria: DIC
- **Bleeding at ROM with fetal bradycardia**: vasa praevia
- **Previous CS + labour + pain + loss of contractions or fetal heart**: rupture
- **Severe hypertension, headache, proteinuria**: pre-eclampsia with abruption
- **Oliguria** (<30 mL/h): renal failure

RED FLAG

Deliver or refer **now**, not after the workup, if: SI ≥1.4 or SBP <90 despite fluids; ongoing heavy bleeding; tense tender uterus with an abnormal or absent fetal heart; bleeding at membrane rupture with fetal bradycardia (vasa praevia); signs of uterine rupture; blood that does not clot.

Thinking like a clinician

- **The key question: pain and uterine tone.** Painless + soft = praevia. Pain + hard = abruption. Pain + shock + fetal parts easily felt + stopped contractions = rupture. Painless, sudden bleeding at ROM + fetal distress with a well mother = vasa praevia.
- **"Shock out of proportion"** = concealed blood (abruption or rupture). Treat the mother's physiology, not the pad count.
- **Abruption and praevia can coexist.** Don't stop thinking once you find one.
- **In labour, a bloody show is normal**: small amount, mixed with mucus, with cervical change. Heavy fresh bleeding is not a show.
- **Every APH at a HC III is a referral** to a CEmONC facility unless it is trivial spotting from a clearly visible benign cervical lesion.
- **Always offer anti-D** to an Rh-negative woman and **steroids** if <34 weeks and delivery is likely.
- **Pre-eclampsia can be masked**: a shocked woman with abruption may have a "normal" BP that is actually high for her. Check urine for protein.
- **Don't forget the cervix**: once praevia is excluded, a woman with recurrent spotting needs a speculum examination. **Cervical cancer in pregnancy** is not rare in Uganda.
- **Common student mistakes**: doing a digital VE; being reassured by a normal BP; forgetting the Rhesus status; forgetting the risk of **PPH after delivery** in both praevia and abruption; tocolysing a suspected abruption.

EXAM FAVOURITE

"A woman at 34 weeks presents with painless vaginal bleeding. What will you NOT do?" **Digital vaginal examination.** Then: "What will you do?" Resuscitate, assess mother and fetus, ultrasound for placental site, group and cross-match, and expectant or active management according to gestation and severity.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Placenta praevia	Painless, recurrent, bright-red bleeding; soft non-tender uterus; high head or malpresentation; FHR usually normal; shock proportional to visible loss	**Ultrasound** (transvaginal is safe and most accurate)
Placental abruption	Painful, dark or bright bleeding (may be scant); tense tender uterus; fetal distress or death; shock out of proportion; pre-eclampsia or trauma	Clinical diagnosis; ultrasound often **misses** it (normal scan doesn't exclude it); CTG; clotting
Uterine rupture	In labour, previous CS or obstruction; sudden pain, cessation of contractions, recession of presenting part, easily felt fetal parts, shock, haematuria, absent FHR	Clinical; laparotomy (don't delay for scan)
Vasa praevia	Bleeding at ROM, mother stable, **acute fetal bradycardia or sinusoidal CTG**	Clinical; antenatal transvaginal colour Doppler; Apt/Kleihauer test on vaginal blood (don't delay delivery)
Bloody show	Small, mucoid, with contractions and cervical change	Clinical (speculum/VE once praevia excluded)
Cervical lesions (ectropion, polyp, cervicitis, **cancer**)	Postcoital spotting, discharge; lesion on speculum; uterus normal	Speculum, VIA, Pap smear or **biopsy** (cervical cancer)
Vaginal trauma / infection / foreign body	History (assault, coitus); lesion or discharge on speculum	Speculum; swabs
Vulvovaginal varicosities	Visible varices	Inspection
Non-genital (haematuria, haemorrhoids)	Blood in urine or stool	Urinalysis, rectal inspection

Praevia vs abruption, the classic viva table

Feature	Praevia	Abruption
Pain	Painless	Painful
Onset	Insidious, recurrent, "warning bleed"	Sudden
Blood	Bright red, fresh	Dark (may be bright), may be concealed
Shock	Proportional to visible loss	**Out of proportion**
Uterus	Soft, non-tender	Tense, tender, **woody**
Presenting part	High; malpresentation	Normal; may be engaged
Fetal heart	Usually normal	Distress or absent
Associated	Previous CS, multiparity	Hypertension, trauma
Coagulopathy	Rare	Common in severe cases

8. Clinical Workup

Every APH is assessed in the same order: **mother first, then fetus, then cause**, and always **without a digital VE** until the placenta has been located. Key principles:

1. **Ultrasound before any digital VE.** It answers the most important question (where is the placenta?) and confirms fetal life. Transvaginal scanning is safe in praevia.
2. **A normal scan does not exclude abruption.** Retroplacental clots are often invisible; abruption is diagnosed clinically.
3. **Serial, not single, tests in abruption**: repeat the 20-minute clotting test every 2–4 h, re-measure the fundal height, and follow Hb and urine output hourly to detect ongoing concealed loss and DIC.
4. **Don't delay delivery for a test** when the mother is shocked or the fetus is distressed (e.g. Apt test in suspected vasa praevia).

Bedside

Test	How / why	Expected finding and what it changes	Available at
Shock index, pad count / weighing (1 g ≈ 1 mL)	Every set of observations	Severity; SI out of proportion to visible loss = concealed bleeding	All levels
Fetal heart: Doppler/Pinard, then **CTG** if viable and available	Continuous if abruption suspected	Distress, bradycardia; **sinusoidal pattern** suggests fetal anaemia (vasa praevia) → timing and urgency of delivery	Pinard/Doppler all levels; CTG at hospitals (many HC IVs have Doppler only)
Bedside ultrasound (skilled operator)	Placental site, FH, lie, liquor	Excludes praevia so you can examine; confirms viability; retroplacental clot only sometimes seen	HC IV and above with an operator
HemoCue / point-of-care Hb	Baseline	Will fall further with concealed loss	HC III and above
20-minute whole blood clotting test	2 mL in a dry glass tube	No clot at 20 min = DIC; in abruption **repeat every 2–4 h** → blood products	All levels
Urine dipstick (protein) and BP	Pre-eclampsia as the cause of abruption	Proteinuria → $MgSO_4$, BP control	All levels
Urine output via catheter	Hourly	<30 mL/h = hypovolaemia or AKI	All levels
Apt (alkali denaturation) test on vaginal blood	Only if it does not delay delivery	Fetal haemoglobin = vasa praevia	Hospital labs Verify *availability in Uganda*
Malaria RDT if febrile	Common co-morbidity in Uganda	Positive → treat	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC	All	Hb; platelets <100 × 10^9/L (DIC, HELLP)	Transfusion; iron; platelet transfusion if low and bleeding	HC IV and above
Group and cross-match; Rhesus	All	Group; 2–4 units (4–6 if PAS)	Blood ready; **anti-D** within 72 h if Rh-negative	HC IV with blood; hospitals
Creatinine, urea, electrolytes; uric acid	Abruption, pre-eclampsia, oliguria	Rising creatinine (above ~80–90 µmol/L in pregnancy)	AKI: strict fluid balance, avoid overload; renal review	Hospitals

Test	When	Expected / abnormal finding	How it changes management	Available at
LFTs	Hypertension, low platelets	Raised transaminases	HELLP; $MgSO_4$, delivery	RRH/NRH, some HC IVs
HVS / swabs	Discharge, cervicitis	Infection	Syndromic or targeted treatment	Hospitals

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Ultrasound, transabdominal	Every APH once stable	Placental site and lower edge; fetal viability, lie, growth, liquor	Praevia confirmed → no VE, admit to CEmONC; placenta high → speculum and look for other causes	HC IV if a sonographer is present; all hospitals
Transvaginal ultrasound (± colour Doppler)	Placenta low on abdominal scan; suspected vasa praevia	Edge **<20 mm** from os = low-lying; covering os = praevia; vessels over the os = vasa praevia	Plans mode and timing of delivery; TVS is **safe** in praevia	RRH/NRH/private mostly
PAS signs on ultrasound (loss of clear zone, lacunae, bladder-wall vascularity)	Praevia or low-lying placenta **over a CS scar**	Suspicious for accreta/increta/percreta	Refer to a tertiary centre (Mulago/Kawempe, Mengo, RRH with specialists); plan CS at 35+0–36+6 weeks with 4–6 units	RRH/NRH; MRI rarely available

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
PT/APTT, fibrinogen	Suspected abruption, IUFD, clotting test abnormal	Prolonged times; **fibrinogen <2 g/L**	FFP, cryoprecipitate; fresh whole blood if components unavailable	RRH/NRH, often not at night
Kleihauer–Betke	Rh-negative after APH ≥20 weeks; trauma	Size of fetomaternal haemorrhage	Extra anti-D (UK practice)	Rarely available in Uganda, where the 1,500 IU vial is used
Cervical cytology, VIA, biopsy	Visible cervical lesion	Dysplasia or cancer	Biopsy is safe in pregnancy; refer to gynae-oncology	VIA at HC III/IV; biopsy and histology at hospitals

UGANDA CONTEXT

At a HC III without ultrasound, the workup stops at the bedside tests: resuscitate, check the fetal heart and BP, **no VE**, first dose of dexamethasone if preterm, and **refer** with an escort, IV running and relatives ready to donate blood. Write in the referral note what you did **not** do (no VE) as well as what you did.

Putting it together

Model one-line summary

> "A 30-year-old G4P3 with two previous CSs at 33 weeks, with her second painless bright-red APH of about 200 mL, now settled; haemodynamically stable (SI 0.74), soft non-tender uterus, breech, FHR 140; ultrasound shows an anterior major praevia with features suspicious for PAS."

Problem list

1. Major placenta praevia, recurrent APH, currently stable.
2. Suspected placenta accreta spectrum (two previous CSs, anterior praevia).
3. Preterm (33 weeks): steroids.
4. Anaemia (Hb 9.4 g/dL): iron, blood reserved.
5. Rh status and anti-D; social: distance from hospital.

Working diagnosis: *placenta praevia (major, anterior) with suspected PAS, for expectant management and planned CS at 35–36 weeks in a tertiary centre.*

Model case presentation: praevia

> "Mrs A, a 30-year-old G4P3 with two previous caesarean sections, at 33 weeks by early scan, admitted with her second episode of painless bright-red vaginal bleeding, about 200 mL, now settled. She is haemodynamically stable (pulse 88, BP 118/72). The uterus is soft and non-tender, breech presentation, FHR 140. Ultrasound confirms an **anterior major placenta praevia** with loss of the retroplacental clear zone, **suspicious for PAS**. Hb 9.4 g/dL, blood group O Rh-positive. She has had 2 doses of dexamethasone. Plan: conservative management in hospital, iron, keep 2 units cross-matched, complete steroids, senior review for PAS, and a **planned CS at 35–36 weeks** with consent for hysterectomy and 4–6 units of blood."

Model case presentation: abruption

> "Mrs B, a 26-year-old para 2 at 36 weeks with pre-eclampsia, presented with sudden severe abdominal pain and dark vaginal bleeding of about 150 mL. She is tachycardic (122) with BP 100/70, the uterus is tense and tender, and **no fetal heart is heard**: **severe (grade 3) placental abruption with IUFD**. Clotting test: clot at 12 min. We have resuscitated with 2 L crystalloid and 2 units of blood, catheterised (urine 20 mL/h), done ARM and started oxytocin. Plan: vaginal delivery, 4-hourly clotting tests, $MgSO_4$ for pre-eclampsia, prepare for PPH, and monitor renal function."

EXAM FAVOURITE

Worked example 1: ultrasound placental localisation.

- Anomaly scan at 20 weeks: "posterior placenta, lower edge 10 mm from internal os." → **Low-lying** (<20 mm). No action now except advice to come in with any bleeding. **Rescan at 32 weeks** (TVS); most will "migrate" as the lower segment forms.
- 32-week TVS: edge now **25 mm** from the os, cephalic, no bleeding → no longer low-lying; **vaginal birth can be planned**.
- If it had been **covering the os** at 32 weeks → **placenta praevia**: no VE, counsel, admit if it bleeds, **planned CS at 36+0–37+0 weeks** (earlier, 35+0–36+6, if PAS is suspected).
- Same report **with a previous CS scar and an anterior placenta** → look actively for **PAS** signs; refer.

EXAM FAVOURITE

Worked example 2: shock index and the concealed bleed. A woman at 35 weeks with pre-eclampsia arrives with 150 mL of dark blood on her pad. Pulse **124**, BP **100/70** → **SI 1.24**. That is far more shock than 150 mL could cause → **concealed abruption**. Her fundal height, marked at **34 cm** on arrival, is **36 cm** an hour later (still bleeding behind the placenta). Fetal heart absent. Clotting test at arrival: **clot at 12 minutes** (normal, but repeat every 2–4 h). At 4 h: **no clot at 20 minutes**; platelets **68 × 10^9/L**, fibrinogen **1.5 g/L** → **DIC**. Plan: transfuse (red cells, FFP, cryoprecipitate or fresh whole blood), ARM + oxytocin for vaginal delivery, $MgSO_4$, urine output hourly, and prepare for PPH.

9. Management

Immediate / emergency

First 5 minutes

1. **Look at the woman and at the pads**: conscious level, pallor, sweating; how much blood, fresh or dark, clots, still flowing?
2. **Vital signs and shock index**: **≥0.9 abnormal, ≥1.4 urgent**.
3. **Call for help** if bleeding is heavy, SI ≥0.9, the uterus is tense or the fetal heart is abnormal: senior midwife and doctor, anaesthetist; alert theatre and the blood bank.
4. **Quick abdominal check**: soft or woody? tender? fundal height (**mark it with a pen**). **Fetal heart** by Doppler or Pinard.
5. **Say it out loud**: "No digital vaginal examination until praevia is excluded." Put it in the notes.
6. **Decide the lane**: massive bleeding or maternal shock → resuscitate and **deliver** (usually emergency CS; vaginal delivery for abruption with IUFD if the mother can be stabilised); a live distressed viable fetus → category 1 CS; stable → full workup (section 8).

ABCDE: massive APH or shock

- **A/B**: airway, **oxygen**, **left lateral tilt**.
- **C**: **two 14–16G cannulae**; send FBC, **cross-match 4–6 units**, clotting test, U&E. **Warmed crystalloid 1 L fast**, then **blood** (O-negative or group-specific if cross-matched blood isn't ready). Catheterise; urine output hourly.
- **D**: conscious level; glucose if drowsy.
- **E**: keep warm; look for the cause (history, abdominal examination, bedside ultrasound). **No VE.**

Then:

1. **Deliver the baby** (and placenta). This is how bleeding stops in praevia and abruption. Usually by **emergency CS** if the mother is unstable, **regardless of fetal viability**, except in abruption with a dead fetus where vaginal delivery can be quick and the mother can be stabilised.
2. Correct coagulopathy (fresh whole blood, FFP, cryoprecipitate, platelets); **TXA 1 g IV** is reasonable in major haemorrhage.
3. Prepare for **PPH** (oxytocin infusion, uterotonics, compression sutures, hysterectomy).
4. HDU care; document; debrief.

Definitive

General principles: every woman with APH

1. **Call for help**: senior midwife, doctor, anaesthetist, and alert theatre and the blood bank.
2. **ABC and resuscitation** as above.
3. **Assess the fetus**: fetal heart (Pinard/Doppler/CTG), gestation, presentation.
4. **Ultrasound** to localise the placenta; **no digital VE**.
5. **Decide**: deliver now or conservative? It depends on (a) **the mother's condition**, (b) **the fetal condition**, (c) **gestation**, (d) **the cause** and (e) **severity**.
6. **Rh-negative**: anti-D immunoglobulin within 72 h (at least **500 IU** in the UK after 20 weeks, with Kleihauer to top up; in Uganda the available vial is often **1500 IU (300 µg) IM**).
7. **Steroids**: if delivery is likely between **24 and 34 weeks** (WHO 2022; units vary at the upper limit), **dexamethasone 6 mg IM every 12 h × 4 doses** (or betamethasone 12 mg IM every 24 h × 2).

8. **Magnesium sulfate for fetal neuroprotection** if delivery is expected before **32 weeks** (4 g IV loading dose over 20 min, then 1 g/h, or the Uganda regimen per protocol).
9. **Anticipate PPH** after delivery: oxytocin infusion ready, TXA, blood.

Placenta praevia

A. Expectant (conservative) management, the principle behind the **Macafee regimen**: prolong the pregnancy to gain fetal maturity, **only if**:

- the mother is **haemodynamically stable** and the bleeding has settled,
- the fetus is **alive, not distressed**, and **preterm** (<37 weeks),
- blood and theatre are available 24/7 (so admit to a CEmONC hospital).

Steps:

1. **Admit** (in Uganda usually until delivery for a major praevia that has bled; women who live far away, lack transport, or are anaemic should stay in hospital).
2. Maintain **Hb ≥10–11 g/dL**: iron + folic acid; transfuse if needed. Keep **2 units cross-matched**.
3. **Steroids** if 24–34 weeks; **anti-D** if Rh-negative.
4. Avoid coitus, VE and heavy exertion; treat constipation.
5. **Tocolysis**: not routine. Some specialists use **nifedipine** short-term in praevia with contractions to allow steroids to work, but **never tocolyse if abruption is suspected** or the mother is bleeding heavily.
6. **Thromboprophylaxis** for prolonged admission (early mobilisation; LMWH if available and not bleeding).
7. Serial ultrasound (placental position, growth); screen for **PAS** if previous CS.
8. Counsel on the need for CS, the risk of PPH and hysterectomy (**consent** for hysterectomy if PAS is suspected).

B. Timing and mode of delivery

Situation	Plan
Heavy bleeding, maternal instability or fetal distress at any gestation	**Emergency CS** (resuscitate first)
Uncomplicated **placenta praevia** (covers the os)	**Elective CS at 36+0–37+0 weeks** (RCOG). Many Ugandan units plan for about 37 weeks
Praevia with **suspected PAS**	Planned CS at **35+0–36+6 weeks** in a tertiary centre with senior obstetrician, anaesthetist, blood (4–6 units), and consent for hysterectomy
Low-lying placenta (edge ≥20 mm from os at a late scan), cephalic, no bleeding	Vaginal birth can be considered
Edge <20 mm from os	CS usually recommended (individualise)

C. At caesarean for praevia

- **Senior surgeon and anaesthetist**; blood cross-matched and in theatre.
- The lower segment is vascular. The surgeon may go through the placenta (quickly!) or around it; deliver the baby fast.
- **Expect PPH** from the poorly contracting lower segment: oxytocin infusion, TXA, **placental-bed haemostatic sutures**, balloon tamponade, uterine artery ligation, B-Lynch, and hysterectomy if needed.
- **PAS**: don't try to remove an adherent placenta. Leave it and proceed to hysterectomy (or conservative management only in specialist centres).

> **UGANDA CONTEXT**
> Where ultrasound isn't available and praevia is suspected in labour, the historical **"double set-up examination"** may still be done in some units: a gentle VE (actually, fornix palpation) **in theatre, with the team scrubbed, blood available and anaesthesia ready**, so that a CS can be done at once if bleeding starts. It is never done on the ward.

Placental abruption

Manage according to **fetal viability, fetal condition and maternal condition**. **Treat the mother first.**

Scenario	Management
Fetus dead (Grade 3)	Aggressive resuscitation (expect large hidden losses: transfuse early, often 4+ units); **check and correct clotting**; aim for **vaginal delivery**: **ARM** (reduces intrauterine pressure and thromboplastin entry, and speeds labour) **+ oxytocin** infusion. Labour is often rapid. **CS only for maternal indications** (uncontrollable bleeding, deterioration despite resuscitation, transverse lie, failure to progress, previous classical scar), because surgery in DIC is dangerous
Fetus alive and distressed, viable	**Immediate CS** after starting resuscitation (unless vaginal delivery is imminent)
Fetus alive, not distressed, ≥37 weeks	**ARM + oxytocin induction** with **continuous fetal monitoring**; low threshold for CS
Fetus alive, not distressed, preterm, mild abruption, mother stable	**Conservative**: admit, steroids, anti-D, close fetal monitoring (CTG, growth scans), Hb, BP; deliver at 37 weeks or earlier if deterioration. **No tocolysis**

Additional abruption points:

- **Monitor for DIC**: clotting test every 2–4 h; platelets; fibrinogen. Use **fresh whole blood**, FFP, cryoprecipitate and platelets as available.
- **Treat pre-eclampsia** if present: **$MgSO_4$** (4 g IV + 10 g IM loading per Uganda regimen, or IV Zuspan), **antihypertensives** (hydralazine, labetalol, nifedipine) once volume-replete. Beware: a low BP in a hypertensive woman may be shock.
- **Strict fluid balance**; urine output >30 mL/h. Oliguria despite adequate fluids suggests ATN, so avoid fluid overload.
- **Anticipate PPH**: active management, oxytocin infusion, TXA; Couvelaire uterus usually still contracts.
- **After delivery**: examine the placenta (retroplacental clot, and the adherent clot's "crater"), histology if possible; check renal function; anti-D; counsel on **recurrence** (about 10%) and the need for early ANC and BP control next time. Low-dose **aspirin** from 12 weeks is appropriate if pre-eclampsia is the underlying cause.
- **Stillbirth care**: bereavement support; lactation suppression if requested (cabergoline 1 mg once, unless hypertensive, where bromocriptine is also avoided; use supportive measures).

Vasa praevia

- **Antenatally diagnosed** (transvaginal colour Doppler): admit from about **30–32 weeks** (especially if far from hospital); **steroids from 32 weeks**; **planned CS at 34–36 weeks** before labour or ROM.
- **Bleeding at ROM with fetal distress**: this is a **category 1 emergency CS** (a matter of minutes). Resuscitate the **baby** at birth with **O-negative blood/volume** (the neonate may need transfusion).
- Don't perform ARM if you can feel vessels in the membranes (pulsatile cord-like structures).

Uterine rupture

1. **Resuscitate** (as above) and cross-match 4 units.
2. **Laparotomy** immediately; don't wait for the BP to recover fully (the bleeding won't stop without surgery).
3. Deliver the baby and placenta (usually dead).
4. **Repair** the rupture if it's a clean tear, the woman is stable and she wants more children (and the tear is not extending into the broad ligament or cervix). Otherwise do a **subtotal (or total) hysterectomy**.
5. **Inspect the bladder and ureters**. If the bladder is repaired, leave a **catheter for 7–14 days**.
6. Antibiotics (ampicillin + gentamicin + metronidazole), transfusion, HDU care.
7. Counsel: if repaired, **future pregnancy is high-risk**: elective CS at about 36–37 weeks before labour; offer contraception or tubal ligation (with **informed consent**, never assumed).

Local causes

- **Cervical ectropion or polyp**: reassure; polypectomy after delivery unless bleeding heavily.
- **Cervicitis/vaginitis**: swab and treat (syndromic management per UCG).
- **Suspicious cervical lesion**: **biopsy** (it is safe in pregnancy); refer to gynae-oncology. Management depends on stage and gestation.
- **Trauma or sexual assault**: document carefully, offer PEP, emergency support, and police/legal referral.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Dexamethasone	6 mg	IM	12-hourly	× 4 doses	If delivery is likely at 24–34 weeks (WHO 2022); first dose before referral
Betamethasone (alternative)	12 mg	IM	24-hourly	× 2 doses	
Anti-D immunoglobulin	500 IU (UK) or 1500 IU/300 µg (common vial)	IM	Once	Within 72 h of bleed	Rh-negative, non-sensitised; Kleihauer to guide extra doses
Magnesium sulfate (neuroprotection)	4 g IV over 20 min (20–30 min) then 1 g/h	IV	Infusion	Until birth or 24 h	If birth is expected before 32 weeks
Magnesium sulfate (pre-eclampsia with abruption)	4 g IV + 10 g IM loading (Uganda Pritchard), or IV Zuspan	IV + IM	Per regimen	24 h after delivery	R-R-U checks (see pre-eclampsia topic)
Oxytocin	10 IU after delivery; induction low-dose titrated infusion	IV/IM	—	—	Anticipate PPH; induction in abruption after ARM
Tranexamic acid	1 g	IV over 10 min	Repeat once after 30 min if bleeding		In major haemorrhage / PPH
Ferrous sulfate + folic acid	200 mg (65 mg elemental iron) + 400 µg–5 mg	Oral	Daily to twice daily	Through admission	Conservative praevia management; anaemia
Nifedipine (tocolysis, specialist only)	Per specialist protocol Verify *tocolytic dose not stated in source*	Oral	Short course	To complete steroids	Praevia with contractions only; **never in suspected abruption** or heavy bleeding

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ampicillin + gentamicin + metronidazole	Standard doses Verify *doses not stated in source*	IV			After laparotomy for rupture
Cabergoline	1 mg	Oral	Once		Lactation suppression after stillbirth if requested; avoid if hypertensive (bromocriptine also avoided)
Hydralazine, labetalol, nifedipine	See pre-eclampsia topic				Only once volume-replete
Blood products: fresh whole blood, FFP, cryoprecipitate, platelets	Per protocol	IV			DIC in abruption; RRH/NRH for components

Referral criteria

Level	What to do	Refer up when
HC III / HC IV without ultrasound, blood or theatre	Resuscitate (IV line, fluids), **no VE**, check fetal heart and BP, first dose of **dexamethasone** if preterm, clear **referral note**	**Every APH** (unless trivial spotting from a clearly benign cervical lesion) → urgently to a CEmONC facility, **accompanied by a health worker**, IV running, blood donors (relatives) identified, NASG for shock if available, call ahead
HC IV / General Hospital (CEmONC)	Ultrasound, blood, emergency CS, laparotomy for rupture, management of abruption with DIC (fresh whole blood)	Suspected PAS; need for blood components, ICU or NICU → RRH/NRH
RRH / NRH (Mulago/Kawempe, Mengo, RRH with specialists)	TVS and PAS assessment, blood components, planned PAS surgery, HDU/ICU, NICU	n/a

Monitoring and follow-up

- **Abruption**: clotting test every 2–4 h, fundal height re-measured, Hb and urine output hourly, continuous fetal monitoring if alive.
- **Expectant praevia**: Hb, serial ultrasound (placental position, growth), 2 units cross-matched; rescan a low-lying placenta at **32 weeks** (TVS) and at 36 weeks if still low.
- **After delivery**: watch for PPH; renal function after severe abruption; anti-D; HDU care after massive haemorrhage.
- **Future pregnancies**: early ANC and BP control after abruption (recurrence about 10%; aspirin from 12 weeks if pre-eclampsia was the cause); elective CS at about 36–37 weeks after a repaired rupture.

10. Complications

Disease-related

Maternal	Fetal / neonatal
Hypovolaemic shock, death	Prematurity (iatrogenic or spontaneous)
DIC (abruption)	Hypoxia, **stillbirth**

Maternal	Fetal / neonatal
PPH (praevia: poor lower-segment contraction; abruption: DIC, Couvelaire)	**Fetal exsanguination** (vasa praevia)
Acute kidney injury, renal cortical necrosis	IUGR (chronic abruption)
Sheehan's syndrome	Neonatal anaemia
Thromboembolism (postpartum, after transfusion or prolonged admission)	Cerebral palsy (hypoxic-ischaemic injury)
Psychological: bereavement, PTSD	Rhesus isoimmunisation (if no anti-D)

Treatment-related

Complication	Cause
Hysterectomy, bladder injury	PAS surgery, rupture repair
Catastrophic haemorrhage	Digital VE in praevia
Anaesthetic complications	Emergency CS, GA in DIC
Transfusion reactions	Blood products
Infection	Surgery, prolonged admission
Fluid overload	Over-resuscitation in ATN
Worse abruption	Tocolysis in suspected abruption

11. Prognosis

- **Praevia**: with expectant management in hospital and planned CS, maternal and fetal outcomes are good; PAS raises the risk of massive haemorrhage and hysterectomy.
- **Abruption**: fetal death usually indicates >50% separation; maternal risk comes from hidden blood loss, DIC (10–20% of severe abruptions) and AKI. **Recurrence about 10%.**
- **Vasa praevia**: perinatal mortality at least 60% when diagnosed in labour; survival over 95% with antenatal diagnosis and planned CS.
- **Rupture**: the baby is usually dead; a repaired uterus makes future pregnancy high-risk.

12. Prevention

- **ANC**: early booking; anomaly scan with **placental localisation**; rescan a low placenta at 32 weeks.
- **Abruption**: BP control and aspirin for women at risk of pre-eclampsia; road safety and screening for domestic violence.
- **PAS and praevia**: avoid unnecessary primary caesarean sections; look for PAS in every woman with a previous CS and a low anterior placenta.
- **Rupture**: partograph and timely CS for obstructed labour; careful use of oxytocin and misoprostol, especially with a scarred uterus; discourage herbal uterotonics.
- **Rhesus isoimmunisation**: anti-D after every APH in Rh-negative women.
- **Cervical cancer**: VIA/Pap screening.

13. Special Populations

- **Previous CS**: praevia + scar → PAS; labour + scar → rupture.

- **Pre-eclampsia**: the commonest cause of abruption; a "normal" BP may be shock; treat with $MgSO_4$ and antihypertensives once volume-replete.
- **Rh-negative**: anti-D within 72 h; Kleihauer where available.
- **Preterm (<34 weeks)**: steroids; $MgSO_4$ neuroprotection <32 weeks.
- **Twins**: praevia risk; abruption after delivery of the first twin.
- **IUFD with abruption**: high DIC risk; aim for vaginal delivery.

14. Key Points

High-yield points

- No digital VE until praevia is excluded.
- Painless + soft uterus = praevia; pain + woody uterus = abruption; pain + shock + easily felt fetal parts + stopped contractions = rupture; bleeding at ROM + fetal bradycardia = vasa praevia.
- Shock out of proportion = concealed abruption or rupture: shock index, mark the fundus.
- A normal scan does not exclude abruption; repeat the clotting test every 2–4 h.
- Abruption with IUFD: resuscitate, correct clotting, ARM + oxytocin.
- Planned CS: praevia 36+0–37+0 weeks; PAS 35+0–36+6 weeks with 4–6 units and hysterectomy consent.
- Anti-D within 72 h; steroids if delivery likely before 34 weeks; $MgSO_4$ <32 weeks.
- Anticipate PPH after both praevia and abruption.
- Every APH at a HC III is a referral.

Numbers & doses

Item	Value
APH gestation threshold	**≥28 weeks** (Uganda); **≥24+0 weeks** (RCOG). Check your unit's definition
RCOG severity	Spotting; minor **<50 mL**; major **50–1000 mL**, no shock; massive **>1000 mL and/or shock**
Low-lying placenta	Edge **<20 mm** from the internal os (after 16 weeks)
Rescan a low placenta	At **32 weeks** (TVS), and at 36 weeks if still low
Planned CS: praevia	**36+0–37+0 weeks**
Planned CS: suspected PAS	**35+0–36+6 weeks**, 4–6 units, hysterectomy consent
PAS risk with praevia	About **3% / 11% / 40% / >60%** with 0 / 1 / 2 / 3+ previous CS
Frequency	Abruption ~**1%** of pregnancies; praevia ~**1 in 200** at term
Abruption recurrence	About **10%**
Fetal death in abruption	Usually **>50% separation**, maternal loss often **>1.5 L**
DIC in severe abruption	About **10–20%**
Clotting test in abruption	2 mL in a dry glass tube, **no clot at 20 min** = DIC; repeat **every 2–4 h**
Shock index	**≥0.9 abnormal; ≥1.4 urgent**
Cross-match	**2–4 units** (4–6 if PAS or massive bleeding)
Dexamethasone	**6 mg IM 12-hourly × 4** if delivery likely before 34 weeks
$MgSO_4$ neuroprotection	**<32 weeks**: 4 g IV over 20–30 min, then 1 g/h
Anti-D	Within **72 h**: 1,500 IU (300 µg) IM (usual Ugandan vial) or 500 IU + Kleihauer (UK)

Item	Value
TXA	**1 g IV over 10 min**, repeat once after 30 min if bleeding (major haemorrhage/PPH)
Vasa praevia	Fetal blood volume **80–100 mL/kg**; planned CS at **34–36 weeks** if diagnosed antenatally
Urine output	**≥30 mL/h**
Catheter after bladder repair (rupture)	**7–14 days**

Exam traps

- **"Do a VE to assess the cervix."** Never, until praevia is excluded by ultrasound.
- **"The scan is normal, so it isn't an abruption."** Ultrasound misses most retroplacental clots; abruption is clinical.
- **"BP 110/70, so she isn't shocked."** Look at the pulse and shock index; in a pre-eclamptic woman this BP may already be low.
- **"Only 150 mL on the pad."** In abruption the visible loss underestimates the real loss; mark and re-measure the fundus.
- **"Tocolyse the abruption to gain time for steroids."** No tocolysis in suspected abruption or heavy bleeding.
- **"Abruption with IUFD needs an urgent CS."** Usually no: resuscitate, correct clotting and aim for vaginal delivery (ARM + oxytocin); CS for maternal indications.
- **"Couvelaire uterus → hysterectomy."** No; it usually contracts.
- **Forgetting anti-D, steroids and the high risk of PPH** after both praevia and abruption.

Questions seniors ask

Viva questions

1. *What are the causes of APH?* Placental (praevia, abruption, vasa praevia), uterine (rupture), local (cervical/vaginal lesions, infection, trauma, cancer), and unexplained.
2. *How do you differentiate praevia from abruption clinically?* See the praevia vs abruption table (section 7).
3. *Why no VE?* It may separate more placenta and cause torrential bleeding.
4. *What is a Couvelaire uterus?* Extravasation of blood into the myometrium in abruption. The uterus looks bruised; it is **not** by itself an indication for hysterectomy.
5. *Why does abruption cause DIC?* Release of thromboplastin (tissue factor) from the damaged decidua and clot, activating the coagulation cascade and consuming factors.
6. *Management of abruption with IUFD?* Resuscitate, correct clotting, ARM + oxytocin for vaginal delivery; CS only for maternal indications.
7. *When is a low-lying placenta rechecked?* If it is low-lying or praevia at the anomaly scan, repeat at **32 weeks** (and at 36 weeks if still low), by transvaginal scan.
8. *Risk of PAS?* Rises with number of previous CSs when praevia is present (about 3% with none, 11% after 1, 40% after 2, over 60% after 3 or more).

On the ward round

- **What will you NOT do in this woman with APH, and why?** A digital VE, because a finger through the os into a praevia can cause torrential bleeding.
- **How do you tell praevia from abruption at the bedside?** Pain and uterine tone: painless with a soft uterus and a high or malpresenting part = praevia; painful, tense, tender uterus with fetal distress and shock out of proportion = abruption.

- **Her pad shows 150 mL but her pulse is 124. What is going on?** Concealed haemorrhage, most likely abruption (or rupture). Resuscitate, cross-match, clotting test, mark the fundal height and prepare to deliver.
- **What is a low-lying placenta, and what will you do about it at 20 weeks?** Edge less than 20 mm from the internal os; advise her to come in with any bleeding and rescan at 32 weeks by TVS.
- **When would you deliver a woman with placenta praevia?** Immediately if bleeding is heavy or she or the fetus is compromised; otherwise a planned CS at 36+0–37+0 weeks (35+0–36+6 if PAS is suspected).
- **Why do you repeat the clotting test in abruption?** DIC can develop over hours, especially with fetal death; repeat every 2–4 h and correct with blood products or fresh whole blood.
- **What must every Rh-negative woman with APH receive?** Anti-D within 72 hours (1,500 IU IM where that vial is stocked), plus a Kleihauer where available.
- **What would make you suspect vasa praevia?** Bleeding at membrane rupture with a well mother and sudden fetal bradycardia or a sinusoidal trace; deliver by immediate CS.

Mnemonics

- **"Praevia: Painless, Placenta low, Presenting part high."**
- **"Abruption: Acute pain, Angry (woody) uterus, Absent FHR, Anuria, and coAgulopathy."**
- **"Vasa praevia: ROM → bleed → brady."**
- **Rupture: "SHOCKED"**: **S**car or obstructed labour, **H**aematuria, **O**utlines of fetus easily felt, **C**ontractions cease, **K**nocked-back (receding) presenting part, **E**xcruciating pain then relief, **D**isappearing fetal heart.
- **"VE = Very Evil"** until praevia is excluded.

OSCE tips

- On the APH station, **say "I will not do a digital VE until praevia is excluded by ultrasound"** early, and examiners will relax.
- Palpate for **tone and tenderness** and **mark the fundal height**.
- Always mention **Rhesus/anti-D, steroids** and **anticipating PPH**.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define antepartum haemorrhage (Uganda vs RCOG).	Bleeding from or into the genital tract **after the period of viability** and before the birth of the baby: from **28 weeks** in Uganda, from **24+0 weeks** in RCOG guidance.
2	The three life-threatening causes of APH.	**Placenta praevia**, **abruptio placentae** and **uterine rupture**. Then vasa praevia and local causes.
3	The golden rule of APH examination.	**No digital vaginal examination until placenta praevia is excluded** (ultrasound, or in theatre with everything ready for CS).
4	Classic features of placenta praevia.	**Painless, causeless, recurrent bright-red bleeding**; soft non-tender uterus; **high presenting part / malpresentation**; fetal heart usually normal; shock proportional to visible loss.
5	Classic features of placental abruption.	**Painful bleeding** (may be scant or concealed), **tense, tender 'woody' uterus**, fetal distress or death, **shock out of proportion** to visible loss, often with pre-eclampsia or trauma.
6	Ultrasound definitions: placenta praevia vs low-lying placenta (RCOG 2018).	After 16 weeks: **praevia** = placenta **covers the internal os**; **low-lying** = placental edge **<20 mm** from the internal os.
7	Traditional types I–IV of placenta praevia.	I: in lower segment, not reaching the os. II: reaches the os edge (marginal; **posterior type II is 'dangerous'**). III: covers the closed os partially. IV: completely covers the os (central). III–IV = major.
8	What is a Couvelaire uterus?	Blood extravasated into the myometrium (to the serosa) in severe abruption, giving a bruised blue-purple uterus. **Not by itself an indication for hysterectomy**; it usually contracts.
9	Why does abruption cause DIC?	The damaged decidua and retroplacental clot release **thromboplastin (tissue factor)** into the maternal circulation, triggering consumption of fibrinogen, platelets and clotting factors, then fibrinolysis.
10	Management of abruption with a dead fetus.	Aggressive resuscitation (expect large hidden loss), **check and correct clotting**, **ARM + oxytocin** for **vaginal delivery**; CS only for maternal indications. Prepare for PPH; monitor urine output.
11	Abruption with a live, distressed, viable fetus: management?	Resuscitate the mother and perform an **immediate (category 1) caesarean section**, unless vaginal delivery is imminent.
12	Vasa praevia: the classic triad.	**Rupture of membranes → painless vaginal bleeding → acute fetal distress** (bradycardia or sinusoidal CTG), with a well mother. It is **fetal blood**.
13	Vasa praevia diagnosed antenatally: plan?	Transvaginal colour Doppler diagnosis; consider admission from 30–32 weeks; **steroids from 32 weeks**; **planned CS at 34–36 weeks** before labour/ROM. Survival >95% vs ≥60% mortality if found in labour.
14	Timing of elective CS for uncomplicated placenta praevia and for suspected PAS (RCOG).	Praevia: **36+0–37+0 weeks**. Suspected PAS: **35+0–36+6 weeks**, in a tertiary centre with blood and consent for hysterectomy.
15	Risk of placenta accreta spectrum with praevia by number of previous CSs.	About **3%** (no previous CS), **11%** (1 previous CS), **40%** (2), **over 60%** (3 or more). (Silver 2006 reports these by the number of the current CS: 3%, 11%, 40%, 61% at the first, second, third and fourth CS.)

#	Question	Answer
16	Antenatal corticosteroid regimen (WHO).	**Dexamethasone 6 mg IM every 12 h × 4 doses** (or betamethasone 12 mg IM every 24 h × 2) when preterm birth is likely within 7 days, from 24 to 34+6 weeks.
17	Signs of uterine rupture.	Previous CS or obstructed labour; sudden severe pain (sometimes relief); **cessation of contractions**; **recession of presenting part**; **fetal parts easily felt**; absent FHR; shock; haematuria; Bandl's ring before rupture.
18	What is Bandl's ring?	A **pathological retraction ring** between the thick upper and the thinning lower uterine segment in **obstructed labour**, rising towards the umbilicus. It warns of **impending rupture**.
19	RCOG APH severity grades.	**Spotting**; **minor** <50 mL settled; **major** 50–1000 mL without shock; **massive** >1000 mL and/or shock.
20	Why do praevia and abruption both carry a high risk of PPH?	Praevia: the placental bed is in the **lower segment, which has little muscle** and contracts poorly. Abruption: **DIC** and a myometrium infiltrated with blood (Couvelaire).
21	Anti-D in APH.	All **Rh-negative non-sensitised** women with APH should receive **anti-D within 72 h** (500 IU UK after 20 weeks with Kleihauer to guide extra dose; commonly 1500 IU/300 µg vial in Uganda).
22	APH: the first 5 minutes.	Look at the woman and the pads; **vitals and shock index**; call for help if heavy bleeding, SI ≥0.9, tense uterus or abnormal FHR; **left lateral tilt**, two 14–16G cannulae, FBC, **group and cross-match**, **20-min clotting tube**; **mark the fundal height**; **fetal heart**; and **no digital VE**.
23	Why mark the fundal height with a pen in suspected abruption?	A **rising fundal height** on re-measurement means **concealed bleeding** behind the placenta, even when little blood is seen vaginally.
24	Does a normal ultrasound exclude placental abruption?	**No.** Ultrasound misses most retroplacental clots. Abruption is a **clinical** diagnosis (pain, tense tender uterus, fetal distress, shock out of proportion). Use the scan to **exclude praevia** and confirm fetal life.
25	Anomaly scan at 20 weeks: placental edge 10 mm from the internal os. Interpretation and plan?	**Low-lying placenta** (<20 mm). Advise her to come in with any bleeding; **rescan at 32 weeks by TVS** (and 36 weeks if still low). If the edge is ≥20 mm later and cephalic, vaginal birth can be considered.
26	How often should the clotting test be repeated in severe abruption, and what are the lab markers of DIC?	Repeat the **20-min whole blood clotting test every 2–4 h**. DIC: no clot at 20 min, **low platelets**, prolonged PT/APTT, **fibrinogen <2 g/L**. Treat with red cells, FFP, cryoprecipitate, platelets or **fresh whole blood**, and deliver.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 32-year-old G4P3 at 33 weeks, with two previous caesarean sections, presents to a HC IV with a second episode of painless bright-red vaginal bleeding, about 150 mL, now stopped. Pulse 86, BP 118/74. The uterus is soft and non-tender; the fetus is breech with FHR 144. What is the MOST important thing to AVOID?

A Digital vaginal examination
B Group and cross-match
C Intramuscular dexamethasone
D Obstetric ultrasound
E Referral to a hospital with theatre and blood

2. A 26-year-old para 2 at 35 weeks with BP 158/104 mmHg and 3+ proteinuria presents with sudden constant abdominal pain and a small amount of dark vaginal bleeding. Pulse 124, BP now 110/70. The uterus is hard and tender; the fetal heart cannot be heard on Doppler. What is the MOST likely diagnosis?

A Placenta praevia
B Vasa praevia
C Uterine rupture
D Placental abruption with concealed haemorrhage
E Cervical ectropion

3. For the woman in the previous question (severe abruption, fetal death, pulse 124), the bedside clotting test shows a clot at 9 minutes. After resuscitation with 2 L crystalloid and 2 units of blood she is stable. The cervix is 3 cm dilated and the presentation is cephalic. What is the BEST plan for delivery? Advanced

A Immediate caesarean section to deliver the dead fetus
B Amniotomy and oxytocin infusion aiming for vaginal delivery, with continued resuscitation and clotting monitoring
C Wait for spontaneous labour over the next week
D Tocolysis with nifedipine and repeat scan tomorrow
E Misoprostol 800 µg vaginally and discharge home

4. A 28-year-old at 38 weeks in active labour has spontaneous rupture of membranes followed immediately by bright-red vaginal bleeding of about 100 mL. She feels well; pulse 84, BP 120/70. The CTG shows a sudden fetal bradycardia of 70 bpm. What is the MOST appropriate management?

A Give oxytocin to speed up labour
B Perform ultrasound to localise the placenta before deciding
C Apply the NASG and transfuse the mother
D Kleihauer test on the vaginal blood and await the result
E Category 1 emergency caesarean section with neonatal team ready to give volume or blood

5. A 30-year-old para 6 with one previous CS was given 'herbs' in the village and has laboured for 20 hours. She arrives at a HC IV in shock (pulse 136, BP 76/40). She reports that severe pain suddenly eased an hour ago and contractions have stopped. Fetal parts are easily felt under the skin and no fetal heart is heard. Catheter urine is bloody. What is the MOST appropriate management?

A Oxytocin infusion to restart labour
B Vacuum extraction
C Resuscitation and immediate laparotomy
D Observe for 6 hours and repeat ultrasound
E Craniotomy (destructive delivery) through the vagina

6. A 24-year-old at 31 weeks presents with 80 mL of painless vaginal bleeding which has settled. Ultrasound shows a posterior placenta praevia covering the os. Mother and fetus are well. Hb 10.2 g/dL, blood group A Rh-negative, antibody screen negative. Which set of actions is MOST appropriate?

A Discharge home with a review in 4 weeks
B Immediate caesarean section
C Admit, dexamethasone 6 mg IM 12-hourly × 4, anti-D within 72 h, cross-match, iron, and plan CS at 36–37 weeks
D Admit and start nifedipine tocolysis for 4 weeks
E Induce labour with misoprostol once steroids are complete

7. Which ultrasound finding best defines a 'low-lying placenta' according to RCOG (2018) after 16 weeks?

A Placenta covers the internal os completely
B Placenta on the posterior wall of the upper segment
C Placental edge more than 50 mm from the internal os
D Placenta with a succenturiate lobe
E Placental edge less than 20 mm from the internal os without covering it

8. A 34-year-old G5P4 with three previous caesarean sections is found at 28 weeks to have an anterior placenta praevia over the scar, with loss of the clear zone and multiple placental lacunae. What is the MOST appropriate plan? Advanced

A Plan vaginal birth after caesarean at term
B Refer to a tertiary centre for planned CS at about 35–36+6 weeks with senior team, 4–6 units of blood and consent for hysterectomy
C Elective CS at 39 weeks at the HC IV
D Manual removal of the placenta at delivery if it does not separate
E No further action as the placenta will migrate upwards

9. A woman at 36 weeks was knocked by a boda-boda 3 hours ago. She has mild abdominal pain and slight spotting. Vital signs are normal, the uterus is slightly tender, and the CTG is normal. Ultrasound shows a normal fundal placenta with no visible retroplacental clot. What is the MOST appropriate next step? Advanced

A Discharge home because the ultrasound is normal
B Emergency caesarean section
C Give tocolysis and discharge
D Admit for continuous fetal monitoring and maternal observation for at least 24 hours, anti-D if Rh-negative
E Perform an immediate amniotomy

10. A 38-year-old para 5 at 30 weeks has recurrent small postcoital bleeding. Ultrasound shows a normal fundal placenta. Speculum examination reveals a friable, exophytic, irregular lesion on the cervix. What is the MOST appropriate next step?

A Cervical punch biopsy
B Reassure; it is a normal pregnancy ectropion
C Cone biopsy immediately
D Termination of pregnancy
E Wait until 6 weeks after delivery to investigate

11. Which statement BEST explains why placenta praevia bleeding is typically painless and recurrent?

A The placenta is abnormally adherent and does not separate
B It is fetal blood, which does not irritate the uterus
C Hypertension causes spiral artery rupture
D Retroplacental clot causes uterine irritability
E Formation and stretching of the lower segment shears the inelastic placenta off its attachment, without a retroplacental clot

12. Four hours after delivery of a stillbirth due to severe abruption, a woman at a regional referral hospital has passed only 40 mL of urine in 3 hours despite 3 L of fluid and 3 units of blood. BP 118/76, pulse 94, JVP not raised, lungs clear. Creatinine has risen from 70 to 210 µmol/L. What is the MOST likely explanation? Advanced

A Acute tubular necrosis secondary to hypovolaemia and DIC
B Blocked urinary catheter
C Ongoing hypovolaemia
D Normal postpartum physiology
E Ureteric injury

13. A 25-year-old primigravida at 33 weeks presents with mild abdominal pain and a small painful bleed. The uterus is mildly tender between contractions; the CTG is reassuring; she is stable. Ultrasound shows a fundal placenta. She lives 5 km from the hospital. What is the BEST management? Advanced

A Immediate caesarean section
B Discharge with an appointment in 2 weeks
C Admit; dexamethasone; anti-D if Rh-negative; close maternal and fetal monitoring; no tocolysis; plan delivery by 37 weeks or earlier if deterioration
D Nifedipine tocolysis until 37 weeks
E Induce labour today with misoprostol

14. A woman at 36 weeks with a known type IV placenta praevia starts bleeding heavily at a regional referral hospital: pulse 128, BP 84/50, fresh blood soaking the bed. The fetal heart is 150. What is the BEST management?

A Continue conservative management and transfuse
B Vaginal delivery with oxytocin augmentation
C Speculum examination to find the source
D Resuscitate and perform an emergency caesarean section regardless of gestation
E Tocolysis with nifedipine

15. A woman at 34 weeks presents with a small APH. Her pulse is 88/min and BP 112/70 mmHg. The uterus is soft, non-tender and there is a transverse lie. At your HC III there is no ultrasound machine, theatre or blood. What should you do?

A Perform a gentle digital VE to rule out praevia before deciding

B Secure IV access, check fetal heart and BP, give dexamethasone, and refer urgently with an escort to a CEmONC facility

C Admit at the HC III for bed rest until 37 weeks

D External cephalic version to correct the lie

E Rupture the membranes to stop the bleeding

Answers & explanations

1. A · Digital vaginal examination

A is right: this is placenta praevia until proven otherwise (painless, recurrent, soft uterus, malpresentation, previous CS). A **digital VE** may provoke catastrophic haemorrhage. B, C, D and E are all appropriate: cross-match blood, give steroids (she is <34+6 weeks), scan to localise the placenta (and look for **PAS** given two previous CSs), and refer to a CEmONC facility.

2. D · Placental abruption with concealed haemorrhage

D is right: pre-eclampsia, sudden pain, a woody-hard tender uterus, **shock out of proportion to the small visible loss** (a 'normal' BP of 110/70 in a previously hypertensive woman with tachycardia is shock) and fetal death = **severe concealed abruption**. A is wrong: praevia is painless with a soft uterus. B is wrong: vasa praevia follows ROM, the mother is well and the bleeding is fetal. C is wrong: rupture usually occurs in labour with a scar or obstruction and easily palpable fetal parts. E is wrong: ectropion doesn't cause pain, shock or fetal death.

3. B · Amniotomy and oxytocin infusion aiming for vaginal delivery, with continued resuscitation and clotting monitoring

B is right: with a **dead fetus** and a stabilised mother, the aim is **vaginal delivery**. ARM reduces intrauterine pressure and thromboplastin entry and speeds labour; oxytocin augments it. Keep transfusing, check clotting 2–4-hourly and prepare for PPH. A is wrong: CS in a woman at risk of DIC carries a high surgical risk and is reserved for **maternal indications**. C is wrong: delay increases the risk of DIC. D is wrong: **tocolysis is contraindicated** in abruption. E is wrong: she needs inpatient care and misoprostol is not ideal once labour has started.

4. E · Category 1 emergency caesarean section with neonatal team ready to give volume or blood

E is right: bleeding at ROM with acute fetal bradycardia and a well mother is **vasa praevia**. The baby is losing its own blood (total volume about 250–300 mL) and may die within minutes. Deliver immediately and be ready to resuscitate and transfuse the neonate. A is wrong: it delays delivery. B and D are wrong: tests waste precious minutes (an Apt/Kleihauer should never delay delivery). C is wrong: the mother is not the one bleeding significantly.

5. C · Resuscitation and immediate laparotomy

C is right: this is **uterine rupture** (previous CS, prolonged/obstructed labour, herbal uterotonics, sudden relief of pain, cessation of contractions, easily felt fetal parts, haematuria, shock). Resuscitate and do an **immediate laparotomy** (repair or hysterectomy, and check the bladder). A is dangerous: oxytocin worsens the rupture. B and E are wrong: the fetus is no longer in the birth canal, and vaginal procedures delay life-saving surgery. D is wrong: delay means death from haemorrhage.

6. C · Admit, dexamethasone 6 mg IM 12-hourly × 4, anti-D within 72 h, cross-match, iron, and plan CS at 36–37 weeks

C is right: stable preterm praevia calls for **expectant (Macafee-style) management** in a facility with blood and theatre, **steroids** (<34+6 weeks), **anti-D** (Rh-negative), blood available, anaemia correction, and **planned CS at 36+0–37+0 weeks**. A is wrong: after a bleed, a major praevia needs close supervision; in Uganda most are kept in hospital because of distance and transport. B is wrong: there is no maternal or fetal indication to deliver at 31 weeks. D is wrong: long-term tocolysis is not recommended. E is wrong: **vaginal delivery is contraindicated** with a praevia covering the os.

7. E · Placental edge less than 20 mm from the internal os without covering it

E is right: **low-lying** = edge <20 mm from the os; **praevia** = covering the os (A). B is a normal site. C is well clear of the os. D is a risk factor for vasa praevia and retained lobe, not a definition of low-lying.

8. B · Refer to a tertiary centre for planned CS at about 35–36+6 weeks with senior team, 4–6 units of blood and consent for hysterectomy

B is right: praevia + three previous CSs + ultrasound signs = **placenta accreta spectrum** (risk over 60% with three previous CSs). Delivery should be **planned** in a tertiary centre at **35+0–36+6 weeks** with a senior obstetrician and anaesthetist, blood and consent for hysterectomy. A is wrong: vaginal birth is contraindicated. C is wrong: waiting until 39 weeks risks emergency bleeding in labour, and a HC IV lacks the resources. D is wrong: attempting to remove an accreta causes massive haemorrhage. E is wrong: 'migration' occurs mainly with low-lying placentas in mid-pregnancy, not an anterior praevia over a scar at 28 weeks with PAS signs.

9. D · Admit for continuous fetal monitoring and maternal observation for at least 24 hours, anti-D if Rh-negative

D is right: after trauma, **abruption can evolve over hours** and **ultrasound misses most abruptions**, so a normal scan does not exclude it. Admit for monitoring (CTG and maternal observations), check Rh status and give anti-D, and consider a Kleihauer. A is wrong: false reassurance from the scan. B is wrong: there is no current maternal or fetal indication. C is wrong: tocolysis is contraindicated in suspected abruption. E is wrong: there is no indication for delivery with a normal CTG and a stable mother at this point.

10. A · Cervical punch biopsy

A is right: a suspicious cervical lesion in pregnancy must be **biopsied** (a punch biopsy is safe in pregnancy) to exclude **cervical cancer**, which is common in Uganda. B is wrong: ectropion is smooth and regular. C is wrong: a cone biopsy in pregnancy causes haemorrhage and preterm birth, and is reserved for special indications. D is wrong: management depends on the histology, stage and gestation. E is wrong: delay can allow progression.

11. E · Formation and stretching of the lower segment shears the inelastic placenta off its attachment, without a retroplacental clot

E is right: as the lower segment forms and stretches from the late second trimester, the non-stretching placenta separates, opening **maternal** sinuses. There is no retroplacental clot irritating the myometrium, so it is painless, and it recurs as stretching continues. A describes accreta. B is wrong: praevia bleeding is maternal (vasa praevia is fetal). C and D describe **abruption**.

12. A · Acute tubular necrosis secondary to hypovolaemia and DIC

A is right: severe abruption causes **AKI** (acute tubular necrosis or, in extreme cases, cortical necrosis) through hypoperfusion and microthrombi. Now that she is haemodynamically restored, oliguria with a rising creatinine indicates renal injury. Manage with careful fluid balance (avoid overload), monitoring potassium, and renal referral or dialysis if needed. B should always be checked (flush the catheter), but it doesn't explain the rising creatinine. C is wrong: she is now normotensive, not tachycardic, and adequately filled. D is wrong: postpartum diuresis is expected, not oliguria. E is wrong: there was no surgery.

13. C · Admit; dexamethasone; anti-D if Rh-negative; close maternal and fetal monitoring; no tocolysis; plan delivery by 37 weeks or earlier if deterioration

C is right: a **mild abruption** in a **preterm** pregnancy with a stable mother and a reassuring fetus is managed **conservatively in hospital**: steroids, anti-D, CTG and serial growth scans, and delivery at about 37 weeks or earlier if the mother or fetus deteriorates. A and E are wrong: premature delivery without a maternal or fetal indication. B is wrong: an abruption can extend suddenly. D is wrong: **tocolysis is contraindicated** in abruption.

14. D · Resuscitate and perform an emergency caesarean section regardless of gestation

D is right: **massive APH with maternal shock** is an indication for **delivery by emergency CS**. Delivery is how the bleeding is stopped, and the maternal indication overrides fetal considerations. Prepare for PPH from the lower segment. A is wrong: conservative management is only for a stable mother. B is wrong: vaginal delivery is impossible and dangerous with a complete praevia. C is wrong: it wastes time and adds no information. E is wrong: tocolysis in a shocked, bleeding woman is dangerous.

15. B · Secure IV access, check fetal heart and BP, give dexamethasone, and refer urgently with an escort to a CEmONC facility

B is right: painless APH with a soft uterus and malpresentation suggests **placenta praevia**. At a facility with no scan, theatre or blood, resuscitate, give the first steroid dose, and **refer early** with a health worker and referral note, calling ahead. A is wrong: **no digital VE** until praevia is excluded. C is wrong: she needs a facility that can do an emergency CS and transfuse. D is wrong: ECV is contraindicated with APH and praevia. E is wrong: ARM is impossible and dangerous with a praevia.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Painless bleeding at 32 weeks

A 35-year-old G5P4 at 32 weeks presents to Mengo Hospital labour ward at 20:00 with sudden painless bright-red vaginal bleeding that soaked two pads, starting while she was asleep. She had a CS 3 years ago for fetal distress. She has not had an ultrasound this pregnancy.

1. What are your immediate actions?

> Call for help; **ABC**; left lateral position; **two large-bore IV cannulae**; bloods for **FBC, group and cross-match (2–4 units), clotting test, Rh status**; start IV fluids if tachycardic; assess the **amount of bleeding** (look at the pads); check the **fetal heart**. **No digital VE.**

2. What focused history will you take, and why each point?

> - **Pain?** Painless suggests praevia.
> - **Previous bleeds** (a warning bleed?): recurrence is typical of praevia.
> - **Fetal movements**: fetal well-being.
> - **Previous CS**: praevia and **placenta accreta spectrum** risk.
> - **Coitus or trauma**: triggers.
> - **Headache, BP history**: abruption.
> - **Rh group, Hb, HIV** from the ANC card.
> - **Contractions/ROM**: labour or vasa praevia.

3. Examination: pulse 96, BP 116/70, uterus soft and non-tender, fundal height 32 cm, head high and floating, FHR 146 regular. What is your differential and next investigation?

> Most likely **placenta praevia** (painless, soft uterus, high head, previous CS). Differentials: marginal (mild) abruption, vasa praevia (less likely: no ROM, fetus well), local cervical lesion, show. **Next: ultrasound** (transabdominal ± transvaginal) for placental site and to look for **PAS signs** over the scar. Speculum only after praevia is excluded.

4. Scan: anterior major praevia covering the os, no PAS signs; bleeding has stopped. How will you manage her?

> **Expectant management** in hospital: admit; **dexamethasone 6 mg IM 12-hourly × 4**; **anti-D** if Rh-negative; keep **2 units cross-matched**; **iron + folic acid** (keep Hb ≥10–11); no VE or coitus; thromboprophylaxis (mobilise; LMWH if available and not bleeding); repeat scan; counsel on the need for CS and the risk of PPH; plan **elective CS at 36+0–37+0 weeks** by a senior, with blood available; **emergency CS** if heavy bleeding, shock or fetal distress at any stage.

5. At CS she bleeds heavily from the placental bed after delivery of the placenta. What will you do?

> **Oxytocin 10 IU IV slowly + infusion**, **TXA 1 g IV**, bimanual compression, **placental-bed figure-of-eight haemostatic sutures**, uterine balloon tamponade inserted from above, **bilateral uterine artery ligation**, B-Lynch, and **hysterectomy** if bleeding persists. Transfuse, and correct clotting. The lower segment contracts poorly, so anticipate this before starting.

6. How would you present her on the ward round the next day?

> 'Mrs S, a 35-year-old para 5 with one previous CS, now day 1 after emergency/elective CS at 36 weeks for major anterior placenta praevia, complicated by PPH of 1500 mL from the placental bed, controlled with uterotonics, TXA, haemostatic sutures and uterine artery ligation. Transfused 3 units. Now stable, pulse 90, BP 110/70, urine 60 mL/h, uterus firm, lochia normal. Hb 8.6. Plan: iron, thromboprophylaxis, breastfeeding support, family planning counselling, and explain her risks next pregnancy.'

> **Take-away:** Painless APH with a soft uterus and a high head is praevia until proven otherwise: no VE, scan, conservative management if stable and preterm, CS at 36–37 weeks, and always expect PPH from the lower segment.

Case 2: A hard uterus and a silent fetal heart

A 27-year-old para 3 at 34 weeks is referred from a HC III in Luweero to the regional referral hospital with 'bleeding and severe abdominal pain' since morning. She missed several ANC visits. The referral note says BP 170/110 at the HC III, where she received $MgSO_4$ 4 g IV + 10 g IM.

1. On arrival: pulse 132, BP 96/60, pale, restless, a small dark clot on the pad. What is your working diagnosis and your immediate actions?

> **Severe placental abruption with concealed haemorrhage and hypovolaemic shock**, on a background of **pre-eclampsia** (her 'normal' BP is actually a fall from 170/110). Call for help, oxygen, left tilt, **two large-bore IV cannulae**, bloods (**FBC, cross-match 4–6 units, 20-min clotting test, U&E**), warmed crystalloid then **blood early**; catheterise (hourly urine; proteinuria check); continue **$MgSO_4$ maintenance** (monitor reflexes, RR, urine output).

2. Examination: the uterus is tense, woody-hard and tender, fundal height 38 cm; no fetal heart on Doppler or ultrasound. The clotting test shows no clot at 20 minutes. Interpret.

> **Grade 3b abruption**: intrauterine fetal death with **coagulopathy (DIC)**. The fundal height above dates shows **concealed bleeding** (mark the fundus and watch it). Blood loss is probably **well over 1.5 L**. Risk of **AKI, PPH and death**.

3. How will you manage her delivery and coagulopathy?

> 1. Continue resuscitation: **transfuse** (fresh whole blood is ideal where components are unavailable; FFP, cryoprecipitate and platelets at the RRH if available).
> 2. Once stabilising: **ARM** plus **oxytocin infusion** for **vaginal delivery** (usually rapid).
> 3. **CS only** if bleeding is uncontrollable, she deteriorates despite resuscitation, or labour fails, and only after correcting clotting as far as possible.
> 4. Repeat clotting tests every 2–4 h.
> 5. **Antihypertensives** only if BP rises again to ≥160/110 (hydralazine/nifedipine).
> 6. Anti-D if Rh-negative.

4. What will you anticipate after delivery?

> **PPH** (DIC plus Couvelaire uterus): active third stage, oxytocin infusion, **TXA 1 g IV**, misoprostol, blood products. **AKI**: strict input/output, creatinine and potassium; avoid fluid overload (pre-eclampsia plus pulmonary oedema risk). **Eclampsia**: continue $MgSO_4$ for 24 h after delivery. **Sheehan's**. **Infection**. Bereavement support and lactation suppression advice (avoid bromocriptine and cabergoline in hypertension; use supportive measures).

5. What follow-up and counselling will she need?

> Postnatal BP review (antihypertensives weaned), renal function, Hb/iron; **debrief** and bereavement counselling; **family planning** (spacing ≥2 years); next pregnancy: **early ANC**, **low-dose aspirin 75–150 mg from 12 weeks**, calcium supplementation, BP monitoring, deliver at a hospital, and the recurrence risk of abruption (~10%). Review for MPDSR if this was a near-miss.

> **Take-away:** In abruption, visible blood lies: treat the shock and the clotting, deliver vaginally if the fetus is dead, and anticipate PPH, renal failure and eclampsia.

Case 3: Bleeding at membrane rupture

A 30-year-old G2P1 at 39 weeks, conceived by IVF, is in active labour at 5 cm at a private hospital in Kampala. Her anomaly scan report mentions a 'low-lying placenta at 20 weeks' but no follow-up scan was done. Her membranes rupture spontaneously and the midwife notes about 100 mL of fresh blood. Two minutes later the CTG shows a fetal heart of 70 bpm.

1. What is the most likely diagnosis and why?

> **Vasa praevia**: **ROM followed immediately by bleeding and acute fetal bradycardia**, with a well mother. Risk factors: **IVF**, **low-lying placenta in the second trimester** (never re-scanned), possibly a velamentous insertion. The blood is **fetal**.

2. What is your immediate management?

> **Category 1 emergency CS** (decision-to-delivery as fast as possible; aim for under 30 min, ideally much less). Call the neonatal team: be ready for **neonatal resuscitation with volume (normal saline 10 mL/kg) and O-negative blood**. Don't waste time on tests. If the cervix is fully dilated and the head is low, the fastest route may be an instrumental delivery.

3. What differential should you briefly consider, and how would you distinguish it?

> - **Placental abruption**: pain, tense uterus, maternal shock.
> - **Praevia/low-lying placenta bleeding**: maternal blood, fetus usually well initially.
> - **Uterine rupture**: previous scar, pain, maternal shock.
>
> The **Apt test / Kleihauer** on vaginal blood shows fetal haemoglobin, but it must **never delay delivery**.

4. What should have been done antenatally to prevent this?

> A **low-lying placenta at the anomaly scan** should be re-scanned at **32 weeks** (transvaginal), and in women with risk factors (IVF, low-lying, bilobed/succenturiate placenta, velamentous insertion) the **cord insertion** and vessels over the os should be checked with **colour Doppler**. If vasa praevia is diagnosed: **steroids from 32 weeks**, consider admission from 30–32 weeks, and **planned CS at 34–36 weeks** before labour.

> **Take-away:** Bleeding at membrane rupture with sudden fetal distress is vasa praevia: the baby is bleeding, so deliver within minutes and be ready to transfuse the newborn.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.